

MUHAMMAD JUNAID

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PROFESSIONAL SUMMARY

Full-Stack Developer with hands-on experience building end-to-end web platforms using React, Node.js, and MongoDB. Experienced in interactive 3D WebGL rendering, LLM API integration, and automated cloud deployments via Vercel and Render. Focused on writing maintainable code and delivering high-performance digital experiences.

TECHNICAL SKILLS

Frontend & UI Engineering: React.js, JavaScript (ES6+), Three.js, Tailwind CSS, HTML5, CSS3, Redux Toolkit
Backend & API Architecture: Node.js, Express.js, RESTful APIs, JWT, Role-Based Access Control, Postman
Databases & Storage: MongoDB Atlas, Mongoose, Schema Validation
DevOps & Cloud Deployment: Git, GitHub, Vercel, Render
AI & System Integration: Groq API, Gemini API, Python, Voice Pipelines

SOFTWARE ENGINEERING PROJECTS

NotesNest — Full-Stack Note Management Platform

[React.js](#), [Node.js](#), [Express.js](#), [MongoDB Atlas](#), [Tailwind CSS](#), [Render](#), [Vercel](#)

- Architected a full-stack notes platform using React and Express.js, engineering 14+ RESTful endpoints backed by MongoDB Atlas for real-time CRUD operations.
- Secured application state with dual-token authentication (short-lived JWT access tokens and HTTP-only refresh tokens), integrating bcrypt password encryption and automated email verification workflows.
- Developed rich-text notes and to-do lists with search, pagination, pinning, favorites, trash and restore, plus scheduled reminders.

Interactive 3D Virtual Simulation Platform (Undergraduate Capstone)

[Three.js](#), [React.js](#), [Redux Toolkit](#), [Node.js](#), [MongoDB](#)

- Developed a high-performance 3D laboratory simulator in Three.js and React, optimizing mesh geometries and animation loops to sustain a stable 60 FPS in modern browsers.
- Programmed a client-side reaction calculation engine to model dynamic titration curves with <50ms compute overhead, rendering synchronized SVG and canvas visual telemetry.
- Designed normalized MongoDB schemas and backend endpoints to log user experimental parameters, enabling instant multiangle session replay via a student dashboard.

Autonomous Quadruped AI Companion & Web Bridge

[Python](#), [PyWebView](#), [Groq LLM API](#), [ESP32-S2 Mini](#), [C++](#)

- Architected an asynchronous Python companion service connecting desktop clients to Groq LLM APIs, executing real-time voice command parsing and robot gait actuation with <100ms round-trip latency.
- Built a PyWebView-based desktop interface and web bridge for robot control, combining live status monitoring, AI-driven conversation, and motion execution in a single unified interface.

CERTIFICATIONS & HONORS

- Google AI Professional Certificate:** Applied AI workflows, prompt design, and intelligent systems (Coursera).
- Google AI Essentials:** Machine learning fundamentals and generative AI models (Coursera).
- Punjab Educational Endowment Fund (PEEF) Scholarship:** Merit-based academic scholarship awarded for top ranking.

EDUCATION

Government College University Faisalabad (GCUF)

Expected Aug 2027

Bachelor of Science in Computer Science (BS CS) — CGPA: 3.80 / 4.00

Punjab, Pakistan

Relevant Coursework: Database Systems, Distributed Computing, Computer Networks, Software Engineering, Data Structures & Algorithms.